

Copy_of_Project_3_Amazon_Orders_Dataset (2)

February 16, 2026

#Import libraries

```
[ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

#Load Data

```
[ ]: df=pd.read_excel("/content/Amazon 2_Raw.xlsx")
df.head()
```

```
[ ]:      Order ID Order Date  Ship Date      EmailID \
0  CA-2013-138688 2013-06-13 2013-06-17  DarrinVanHuff@gmail.com
1  CA-2011-115812 2011-06-09 2011-06-14  BrosinaHoffman@gmail.com
2  CA-2011-115812 2011-06-09 2011-06-14  BrosinaHoffman@gmail.com
3  CA-2011-115812 2011-06-09 2011-06-14  BrosinaHoffman@gmail.com
4  CA-2011-115812 2011-06-09 2011-06-14  BrosinaHoffman@gmail.com
```

```
      Geography      Category \
0  United States,Los Angeles,California  Labels
1  United States,Los Angeles,California  Furnishings
2  United States,Los Angeles,California  Art
3  United States,Los Angeles,California  Phones
4  United States,Los Angeles,California  Binders
```

```
      Product Name      Sales  Quantity \
0  Self-Adhesive Address Labels for Typewriters b...  14.620      2
1  Eldon Expressions Wood and Plastic Desk Access...  48.860      7
2  Newell 322  7.280      4
3  Mitel 5320 IP Phone VoIP phone  907.152      4
4  DXL Angle-View Binders with Locking Rings by S...  18.504      3
```

```
      Profit
0  6.8714
1  14.1694
2  1.9656
3  90.7152
```

4 5.7825

1 Create Basic KPIs (Mostly important interview question)

#Total Sales

```
[ ]: #Total sales
total_sales= df['Sales'].sum()
print("Total Sales:", total_sales)
```

Total Sales: 725457.8245

#Total profit

Meaning : real after earning after cost

```
[ ]: total_profit = df["Profit"].sum()
print("Total Profit:",total_profit)
```

Total Profit: 108418.4489

#Total Orders

```
[ ]: total_orders=df['Order ID'].nunique()
print("Total Orders:",total_orders)
```

Total Orders: 1611

#TotalQuantity Sold

```
[ ]: total_qty = df["Quantity"].sum()
print("Total Quantity:",total_qty)
```

Total Quantity: 12264

#Step 3 - Business Analysis (Real Company Questions)

1.1 1. Category wise sales

```
[ ]: cat_sales =df.groupby("Category")["Sales"].sum()
print(cat_sales)
```

Category	
Accessories	61114.1160
Appliances	30236.3360
Art	9212.0660
Binders	55961.1130
Bookcases	36004.1235
Chairs	101781.3280
Copiers	49749.2420

```

Envelopes      4118.1000
Fasteners      923.2160
Furnishings    30072.7300
Labels         5078.7260
Machines       42444.1220
Paper          26663.7180
Phones         98684.3520
Storage        70532.8520
Supplies       18127.1220
Tables         84754.5620
Name: Sales, dtype: float64

```

#Profit by category

```
[ ]: cat_profit = df.groupby("Category")['Profit'].sum()
print(cat_profit)
```

```

Category
Accessories    16484.5983
Appliances     8261.2699
Art            2374.0970
Binders        16096.8016
Bookcases     -1646.5117
Chairs         4027.5843
Copiers        19327.2351
Envelopes      1908.7624
Fasteners       275.1907
Furnishings    7641.2704
Labels         2303.1223
Machines       -618.9264
Paper          12119.2364
Phones         9110.7426
Storage        8645.3222
Supplies        626.0465
Tables         1482.6073
Name: Profit, dtype: float64

```

#Top 5 Products

```
[ ]: top_products=df.groupby('Product Name')['Sales'].sum().
    ↪sort_values(ascending=False).head()
print(top_products)
```

```

Product Name
Canon imageCLASS 2200 Advanced Copier
13999.960
High Speed Automatic Electric Letter Opener
13100.240
Global Troy Executive Leather Low-Back Tilter

```

```

10019.600
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind
8134.336
GuestStacker Chair with Chrome Finish Legs
8030.016
Name: Sales, dtype: float64

```

```

[ ]: geo_sales=df.groupby('Geography')['Sales'].sum()
      print(geo_sales)

```

```

Geography
United States,Albuquerque,New Mexico      2220.160
United States,Anaheim,California          7986.870
United States,Antioch,California           19.440
United States,Apple Valley,California     1915.081
United States,Arvada,Colorado              503.400
...
United States,Westminster,California      3826.646
United States,Whittier,California          444.768
United States,Woodland,California          264.662
United States,Yucaipa,California           50.800
United States,Yuma,Arizona                 840.865
Name: Sales, Length: 170, dtype: float64

```

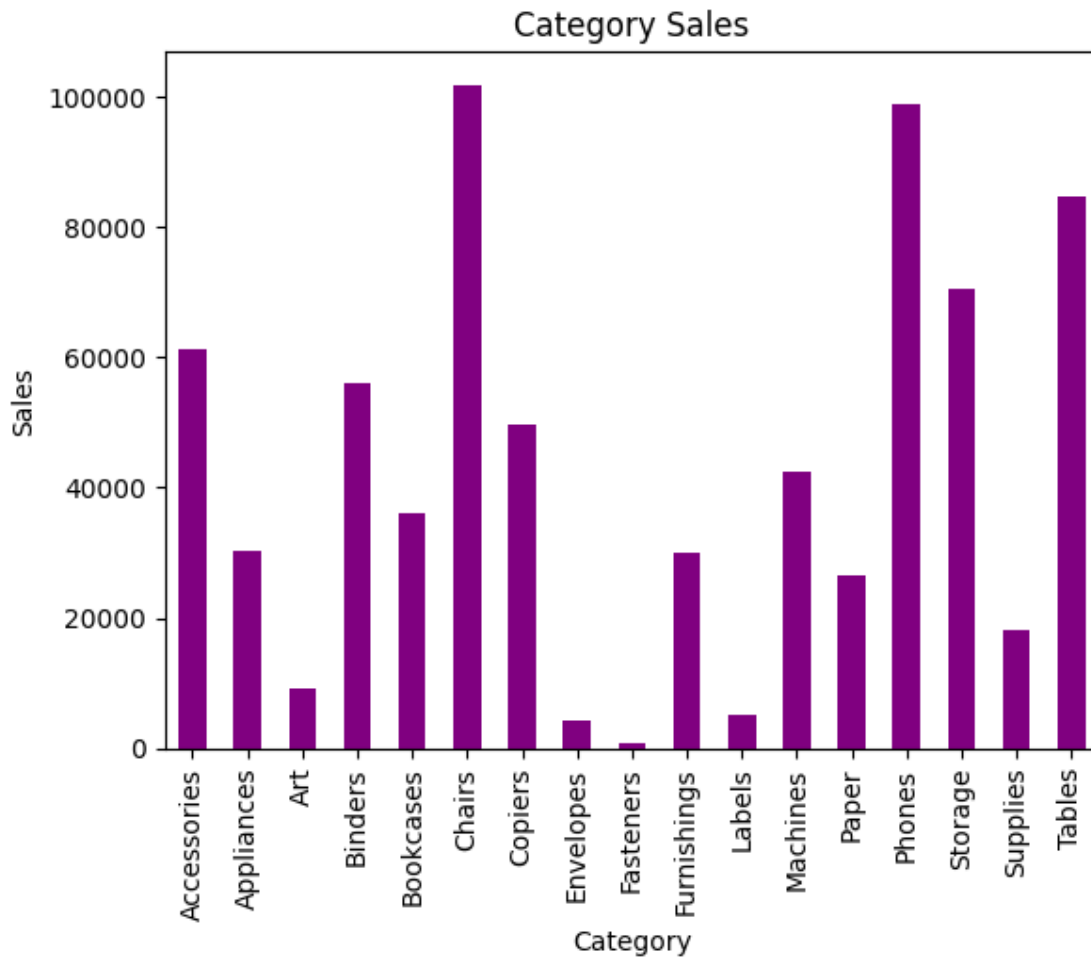
```
#Chart visualization
```

```
##Category sales
```

```

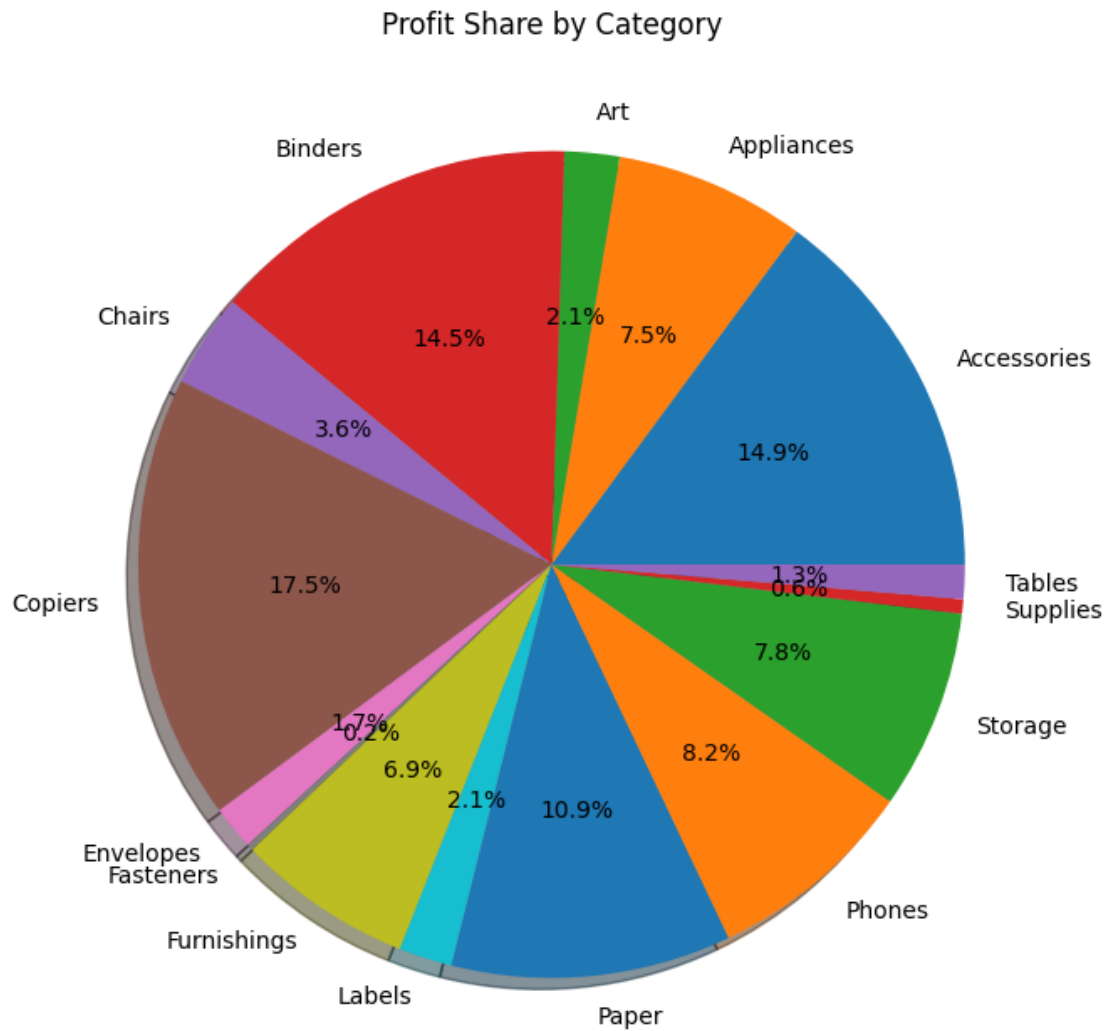
[ ]: cat_sales.plot(kind='bar',color='Purple')
      plt.title("Category Sales")
      plt.xlabel("Category")
      plt.ylabel("Sales")
      plt.show()

```



#chart 2 -Profit Pie Chart

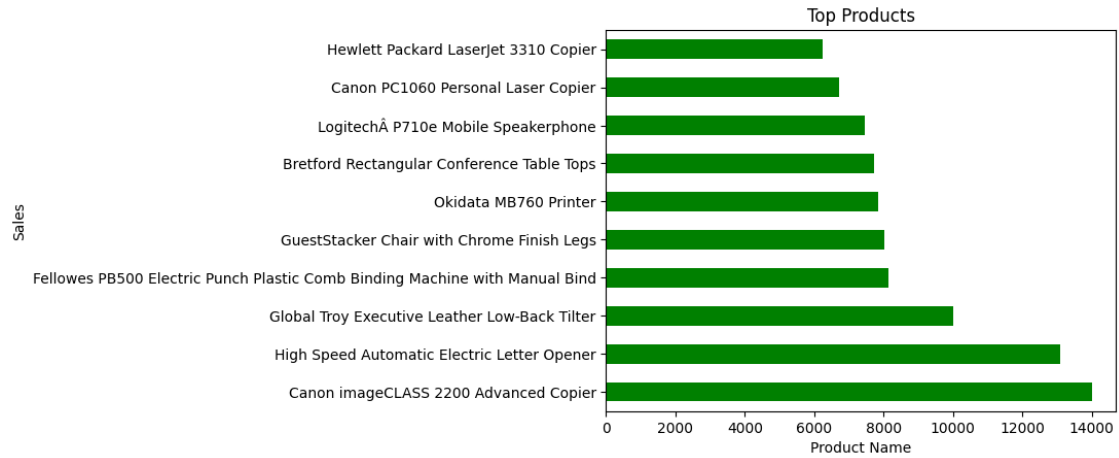
```
[ ]: import matplotlib.pyplot as plt
      #Step 1 grouping the data and sum (Mandatory)
      cat_profit=df.groupby('Category')['Profit'].sum()
      # Filter for positive profits for a pie chart
      cat_profit = cat_profit[cat_profit > 0]
      plt.figure(figsize=(14,8))
      plt.pie(cat_profit, labels=cat_profit.index,autopct='%1.1f%%',shadow=True)
      plt.title('Profit Share by Category')
      plt.show()
```



#Autopct mean show percentage

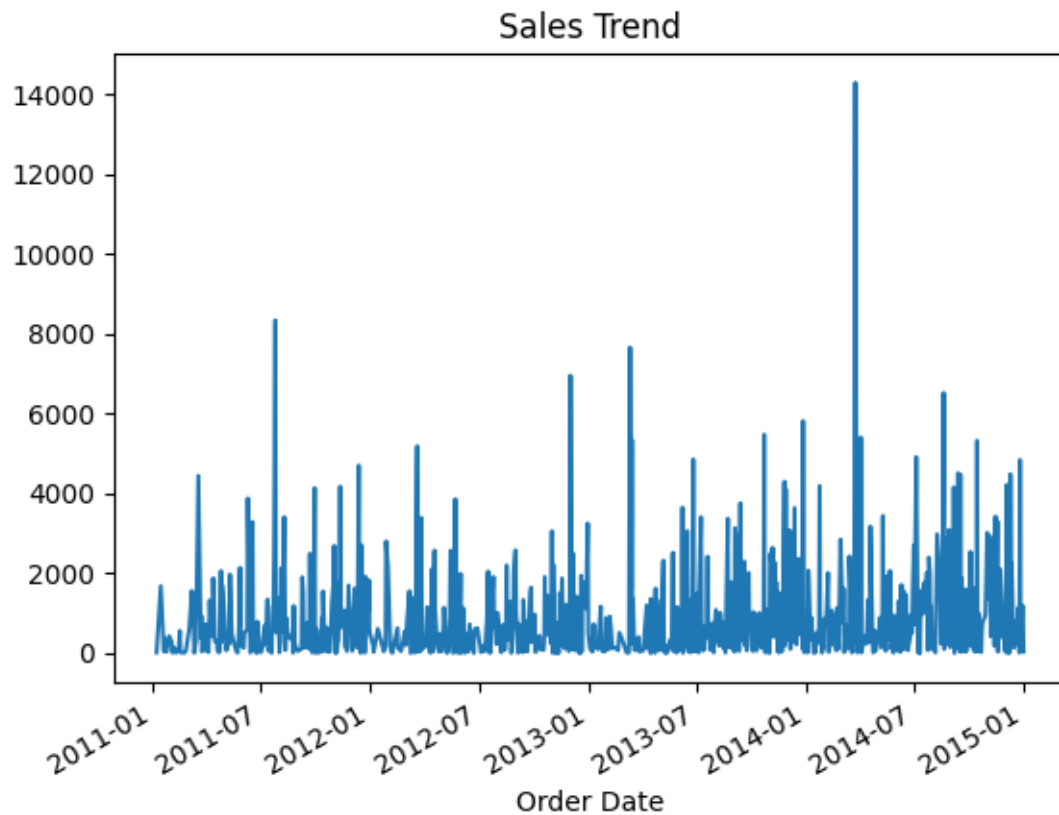
#Chart 3 - Top Products

```
[ ]: top_products.plot(kind = 'barh',color='Green')
plt.title('Top Products')
plt.xlabel('Product Name')
plt.ylabel('Sales')
plt.show()
```



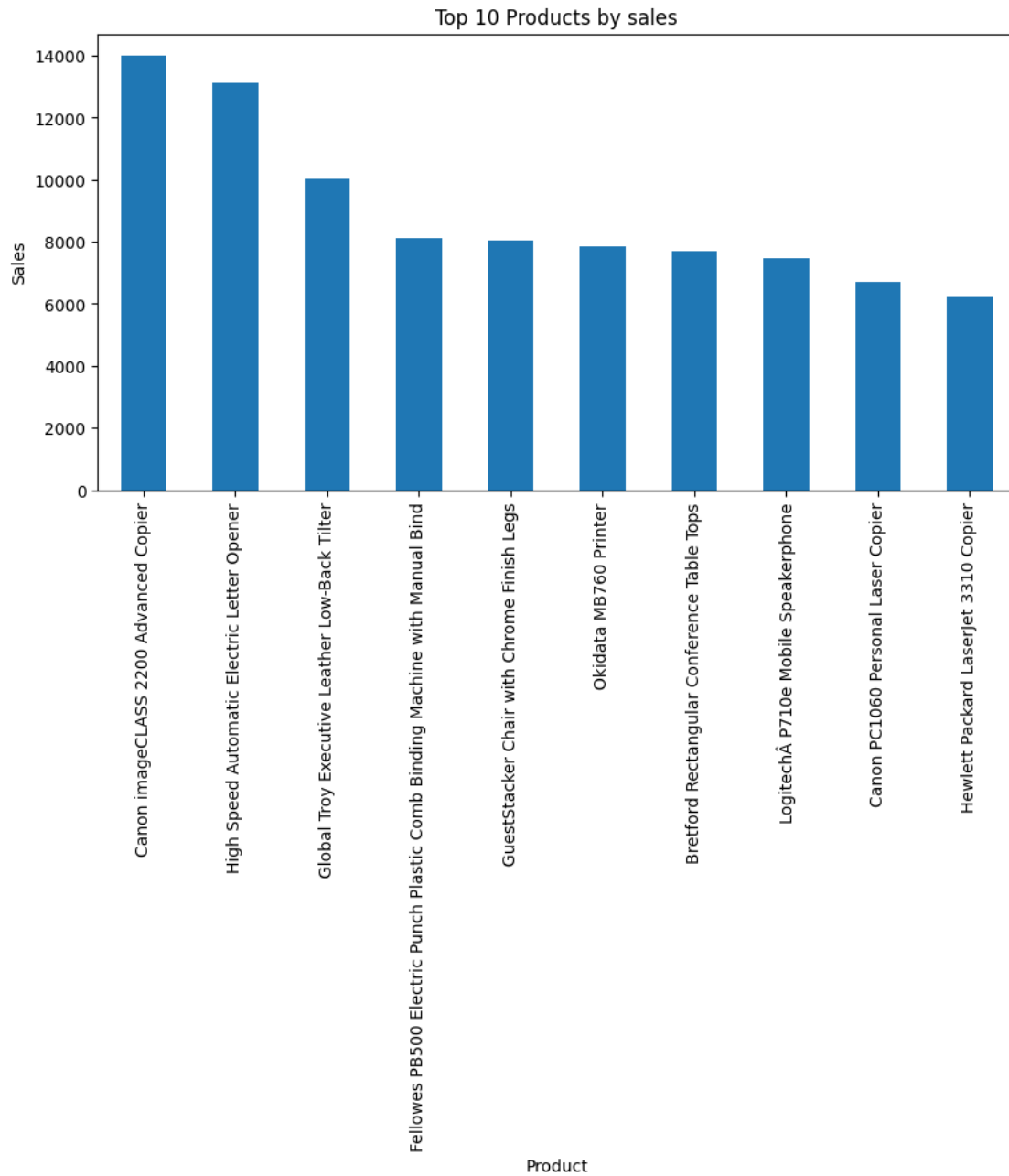
#Chart 4 -Sales Trend by Date

```
[ ]: df['Order Date'] = pd.to_datetime(df['Order Date'])
date_sales =df.groupby ('Order Date')['Sales'].sum()
date_sales.plot()
plt.title("Sales Trend")
plt.show()
```



#Top 10 Products by sales

```
[ ]: top_products = df.groupby('Product Name')['Sales'].sum().  
      ↪sort_values(ascending=False).head(10)  
plt.figure(figsize=(10,5))  
top_products.plot(kind='bar')  
plt.title("Top 10 Products by sales")  
plt.xlabel("Product")  
plt.ylabel("Sales")  
plt.show()
```



#Monthly sales trend

```
[ ]: df['Order Date'] = pd.to_datetime(df['Order Date'])
monthly = df.groupby(df['Order Date'].dt.to_period('M'))
['Sales'].sum()
monthly.plot(figsize=(10,5))
plt.title("Monthly Sales Trend")
plt.ylabel("Sales")
```

```
plt.show()
```

```
#Profit vs Sales
```

```
[ ]: plt.figure(figsize=(6,5))
plt.scatter(df['Sales'],df['Profit'])
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.title("Sales vs Profit Realationship")
plt.show()
```



```
[ ]: 
```

```
[ ]: 
```

```
[ ]: 
```

```
[ ]: 
```

[]:

[]:

[]:

[]:

[]: